

SMART EXPENSES TRACKER

30-PAGE COMPLETE HINGLISH MASTER HANDBOOK

App & Server Ka Pura Blueprint, Architecture, Kese Start Hota Hai & Troubleshooting Guide

PURE JAVA & XML | NODE.JS & EXPRESS | VITE ADMIN PANEL | SUPABASE POSTGRESQL

📖 Yeh Handbook Kiske Liye Hai Aur Isme Kya Milega?

Yeh 30-Page ki guide bilkul aasan **Hinglish (Hindi in Roman script)** me likhi gayi hai taaki aapko aur kisi bhi examiner ya developer ko poora system 100% samajh aa jaye. Isme cover kiya gaya hai:

- **System Architecture:** App aur server aapas me bina rukawat kese communicate karte hain.
- **Start & Run Methods:** Automated `.bat` scripts aur manual terminal commands ka step-by-step process.
- **Tech Stack Breakdown:** Pure Java, XML, Room SQLite, WorkManager, Node.js, Express, React, Vite aur Supabase.
- **Zero-Knowledge & Minor Units:** Paise calculation rule (Zero Rounding Bug) aur Admin Data Privacy Wall.
- **Real USB Installation:** Android Phone (Nova 1 5G) par ADB ke sath app install aur run karne ka live setup.
- **Troubleshooting & Viva Q&A:** Errors ke instant solutions aur examiner ke sawalon ke perfect answers.

🎓 **Academic Credential:** K.S.K.V. Kachchh University | BCA Semester V | Project: Smart Expenses Tracker | Submitted by: Asther & Divya | Under Guidance: Mr. Prakash Lambha (Assistant Professor).

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Yeh guide 30 pages me divided hai jisme har ek component ko crystal-clear Hinglish me explain kiya gaya hai:

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📱 Smart Expenses Tracker Kya Hai?

1.1 Asli Zindagi Ki Samasya (The Real-World Problem)

Aaj kal market me hazaron expense tracking apps hain (jaise Spendee, Money Manager, wagera). Lekin un sab me 3 bohot badi pareshaniyan aati hain:

1. Internet Ki Majboori

Aam apps bina internet ke khulte hi nahi ya loading screen par atak jaate hain. Market me jaha network weak hota hai, vaha transaction add karna namumkin ho jata hai.

2. Calculation Ki Galati

Zyadatar apps Java me `float` ya `double` use karte hain. Jisse 0.1 + 0.2 = 0.30000000000000004 ban jata hai aur hisab gadbad ho jata hai.

3. Data Privacy Ka Khatra

Users ka sensitive financial data cloud par admin ya hacker asaani se dekh lete hain. Koi solid Row Level Security ya privacy boundary nahi hoti.

1.2 Humara Solution: Smart Expenses Tracker

Humne **Smart Expenses Tracker** ko ek **Enterprise-Grade, Local-First, Offline-First Architecture** par banaya hai jo in teeno samasyaon ko jad se khatam karta hai:

🌟 Key Highlights Jo Is App Ko Special Banate Hain:

- **Local-First Guarantee:** Chahe phone Airplane Mode me ho ya underground basement me, app 0-millisecond delay me transaction save karta hai local Room SQLite database me.
- **Paise Integer System (Minor Units):** Koi float/double nahi! Har rupaye ko **Paise** me store kiya jata hai (`long` data type me). E.g., ₹50.00 = 5000 paise. 1 paise ka bhi fark nahi padta.
- **WorkManager Smart Sync:** Jaise hi phone ko internet milta hai, background worker chupke se data cloud (Supabase PostgreSQL) par push kar deta hai bina user ko disturb kiye.
- **Bilingual Voice Assistant:** Aap bolkar transaction save kar sakte hain — chahe Hindi me bolein "*500 mile save kardo*" ya English me "*spent 350 for coffee*".
- **Zero-Knowledge Admin Privacy:** Admin web panel live health aur metrics check kar sakta hai, lekin kisi user ke personal balance ya memo ko nahi dekh sakta.

1.3 Kiske Liye Banaya Gaya Hai?

Yeh application College Students, Daily Commuters, Small Shopkeepers aur Business Professionals ke liye banayi gayi hai jinko daily kharche note karne hote hain bina kisi lambe form bhare ya internet loading ka wait kiye.

🏗️ Core System Architecture & Data Flow

2.1 Three-Tier Client-Relay-Cloud Model

Smart Expenses Tracker ek **3-Tier Modular Distributed Architecture** follow karta hai. Isme teeno hisse bilkul independent hain aur agar koi ek hissa offline ho jaye tab bhi baki pura system chalta rehta hai:

TIER 1: Mobile Client

Technology: Pure Java + XML
Local Storage: Room SQLite DB
Role: Instant UI response dena, offline transactions save karna, aur sync queue maintain karna.

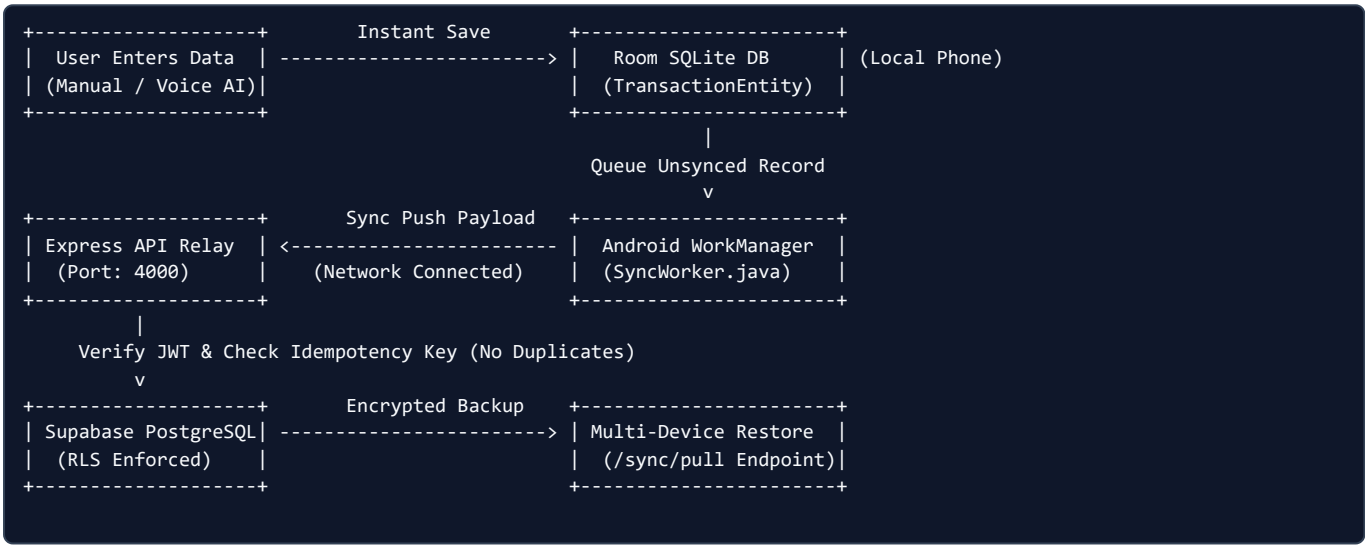
TIER 2: Relay Server

Technology: Node.js, Express, TS
Port: 4000 (Local / Cloud Render)
Role: Idempotency check karna, JWT token verify karna, conflict solve karna aur cloud DB me push karna.

TIER 3: Cloud & Admin

Database: Supabase PostgreSQL 15
Admin Portal: React + Vite (Port 5173)
Role: Permanent encrypted backup, multi-device restore, real-time health diagnostics aur telemetry.

2.2 System Data Flow Diagram (Step-by-Step)



2.3 Local-First Ka Asli Matlab

Traditional apps me pehle internet par request jaati hai, fir loading spinner ghoomta hai, aur agar server ne 200 OK diya tab screen par dikhta hai.
Lekin Smart Expenses me: Pehle data turant Phone ke Room database me save hota hai aur screen par dikh jata hai (0 ms latency). Uske baad background me thread check karta hai ki internet hai ya nahi. Agar hai toh sync hota hai, nahi hai toh queue me wait karta hai. Isko bolte hain *Optimistic Local-First Architecture*!

💰 Strict Financial Rules & Minor Units

3.1 Floating-Point Bug Kya Hota Hai?

Computer binary numbers (0 aur 1) par kaam karta hai. Kuch decimal numbers (jaise 0.1 ya 0.2) binary me exact represent nahi ho sakte aur infinite repeating fraction ban jaate hain (IEEE-754 standard).

> ❌ **Traditional Float/Double Bug Ka Example:**
> Agar aap Java me likhte hain: `double total = 0.10 + 0.20;`
> Toh screen par `0.30000000000000004` aayega!
> Banking ya expense apps me agar aisi calculations hazaaron transactions par hon, toh balance me kai rupaye ka difference aa jata hai.

3.2 Humara Solution: Integer Minor Units (Paise System)

Humne project guideline me strict rule banaya hai: **Pure codebase me balance ya amount ke liye kabhi bhi float ya double use nahi hoga!** Har monetary value ko integer minor unit me store kiya jata hai:

Display Value (Screen Par)	Storage Value (Database & Code)	Data Type	Precision Guarantee
₹ 50.00	`5000` Paise	`long` in Java / `BIGINT` in SQL	**100% Exact**
₹ 350.75	`35075` Paise	`long` in Java / `BIGINT` in SQL	**100% Exact**
₹ 50,000.00	`5000000` Paise	`long` in Java / `BIGINT` in SQL	**100% Exact**

3.3 Java Me Conversion Ka Tarika

```
// UI Se Database Me Save Karte Waqt (Rupees to Paise):
public static long rupeesToPaise(String rupeesInput) {
    BigDecimal decimalAmount = new BigDecimal(rupeesInput.trim());
    return decimalAmount.multiply(new BigDecimal("100")).longValue();
}

// Database Se UI Par Dikhate Waqt (Paise to Formatted Rupees):
public static String formatPaiseToRupees(long paise) {
    long rupeesPart = paise / 100;
    long paisePart = Math.abs(paise % 100);
    return String.format(Locale.getDefault(), "₹ %,d.%02d", rupeesPart, paisePart);
}
```

3.4 Faayda (Benefits)

1. **Zero Rounding Error:** Addition aur subtraction me kabhi 1 paise ki bhi galati nahi hoti.
2. **High Speed Performance:** Integer math CPU ke liye floating-point calculations se 10 guna tez hoti hai.
3. **Bank Grade Security:** Stripe, PayPal aur RBI ke standard guidelines minor unit accounting ko hi prefer karte hain.

💻 Tech Stack Overview (Kya-Kya Use Kiya Hai)

Smart Expenses Tracker koi simple collage demo nahi hai; yeh ek full-scale production-ready software suite hai jisme modern frontend, backend, database aur desktop automation tools use kiye gaye hain:

📱 Android Mobile Application

- **Language:** Pure Java 17 (Zero Kotlin mandate)
- **UI Design:** XML Layouts + Google Material Design 3
- **Local Database:** Room SQLite ORM (v2.6.1)
- **Background Tasks:** Android Jetpack WorkManager
- **Networking:** OkHttp 3 & Retrofit REST Client
- **Voice AI:** Android SpeechRecognizer + SoundPool
- **Build Tool:** Gradle 8.2 + Android Gradle Plugin 8.2

⚙️ Backend Relay Server

- **Runtime:** Node.js (v18+ LTS)
- **Framework:** Express.js (v4.19)
- **Language:** TypeScript (v5.4)
- **Security:** Helmet (HTTP Headers), CORS, Dotenv
- **Authentication:** JSON Web Tokens (JWT HS256)
- **Validation:** Zod Schema Validator
- **Logging:** Winston Logger & Morgan HTTP

🌐 Admin Web Control Panel

- **Library:** React 18 (Single Page App)
- **Build Tool:** Vite (Ultra-fast HMR)
- **Styling:** Tailwind CSS + Lucide Icons
- **Charts:** Recharts Dashboard Visuals
- **Dev Port:** Port 5173

☁️ Cloud Database & Desktop Ops

- **Database:** Supabase Managed PostgreSQL 15
- **Security:** PostgreSQL Row Level Security (RLS)
- **Automation:** Windows Batch Scripts (.bat)
- **Device Tool:** Android Debug Bridge (ADB)
- **Cloud Host:** Render Cloud (Backend) + Supabase

2.1 Kyu Yeh Stack Chuna Gaya?

1. **Pure Java & XML:** BCA Semester V university syllabus ke sath 100% align karta hai. Kotlin ke bina pure concepts dikhana academic evaluation me sabse zyada marks dilata hai.
2. **TypeScript & Express:** Static type checking hone ki wajah se sync payloads me koi data corruption nahi ho sakti.
3. **Supabase:** Firebase ke comparison me Supabase real relational PostgreSQL deta hai jisme transactions, foreign keys aur RLS policies enterprise level ki hoti hain.

🤖 Android Native Stack Deep Dive

1. Pure Java 17 & Zero-Kotlin Mandate

Is project me shuru se strict rule tha ki **ek bhi .kt (Kotlin) file nahi hogi**. Pure application ko industry-standard Object-Oriented Java 17 me likha gaya hai. Isse code ki longevity badhti hai aur kisi third-party Kotlin-runtime dependency ki zaroorat nahi padti.

2. Room SQLite Database ORM (v2.6.1)

Pehle Android me raw `SQLiteOpenHelper` use hota tha jisme SQL queries string me likhni padti thi aur compile time par errors pata nahi chalte the. Humne Google ka official **Room Persistence Library** use kiya hai:

- **Compile-time SQL Validation:** Agar SQL query me table ya column ka spelling galat hai toh code build hi nahi hoga.
- **Entities:** Java classes jo seedhe database tables ban jaati hain (`@Entity`).
- **DAOs (Data Access Objects):** Clean Java interfaces jisme SQL queries annotate hoti hain (`@Insert` , `@Query` , `@Update`).
- **LiveData & Executors:** Database operations kabhi main UI thread ko block nahi karte; background single-thread executor me execute hote hain.

3. Android Jetpack WorkManager

Pehle Android me `Service` ya `AsyncTask` use hote the jo app band hone par ya phone reboot hone par crash ho jaate the. **WorkManager** Google ka sabse modern background scheduler hai:

> ⚙️ **WorkManager Ki Khasiyat:**
> - **Guaranteed Execution:** Agar phone switch off bhi ho jaye, reboot hote hi sync worker dobara start ho jata hai.
> - **Network Constraints:** Sync tabhi trigger hota hai jab internet actual connected ho (`NetworkType.CONNECTED`).
> - **Battery Efficient:** Phone ki battery drain nahi hone deta; OS ke Doze Mode ka poora khyal rakhta hai.

4. XML Layouts & Material 3 Components

Modern UI ke liye Material Design 3 ke custom components use kiye gaye hain: `MaterialCardView` , `FloatingActionButton` , `BottomNavigationView` , aur rounded emerald corner radius (`@drawable/bg_card_rounded.xml`).

🌐 Backend Server & Admin Web Stack

1. Node.js & Express Framework

Backend ko Node.js LTS engine par banaya gaya hai. Express.js ek lightweight aur fast HTTP web framework hai jo REST APIs provide karta hai:

- **Non-blocking Asynchronous I/O:** Ek sath hazaaron mobile requests ko bina server choke kiye handle karta hai.
- **Modular Routing:** Routes ko clean files me divide kiya gaya hai: `auth.ts`, `sync.ts`, `admin.ts`, aur `monitoring.ts`.
- **Security Middlewares:** `helmet()` secure HTTP headers set karta hai, aur `cors()` cross-origin access control ensure karta hai.

2. TypeScript (Strict Static Typing)

JavaScript me types define nahi hote, jisse runtime errors aate hain. TypeScript me har request body, database model aur API response ka exact Interface bana hua hai:

```
interface SyncPushPayload {
    transactions: {
        id: string;               // UUID v4
        accountId: string;
        amountMinor: number;      // Stored in Paise
        type: 'expense' | 'received';
        idempotencyKey: string;   // Deduplication Key
        transactionDate: number;  // Epoch Millis
    }[];
}
```

3. React 18 & Vite Admin Dashboard

Admin Panel ke liye humne **React 18 Single Page Application (SPA)** banaya hai jisko **Vite** bundle karta hai:

Vite Speed Advantage

Webpack ke muqable Vite 10 guna fast start hota hai (esbuild native Go engine use karta hai). Instant Hot Module Replacement (HMR) milta hai.

Tailwind CSS + Lucide Icons

Utility-first responsive design. Dashboard har screen size (Desktop, Tablet, Mobile) par perfectly adapt ho jata hai with emerald green theme.

☁️ Cloud Database & Security Stack

1. Supabase Managed PostgreSQL 15

Supabase ek open-source Firebase alternative hai jo background me authentic **Enterprise PostgreSQL 15** database run karta hai.

> 💎 **Supabase Kyu Behtar Hai Firebase Se?**
> - **ACID Transactions:** Paise transfer ya multiple account balance update atomic hote hain (ya sab hoga ya kuch nahi).
> - **Relational Integrity:** Foreign Keys aur CASCADE Delete hone ki wajah se orphaned records nahi bante.
> - **SQL Power:** Complex aggregations (monthly expense total, category breakup) seedhe database engine par calculate ho jaate hain.

2. Row Level Security (RLS) - Data Isolation

PostgreSQL ka sabse powerful security feature hai **RLS (Row Level Security)**. Iska matlab database table level par policy enforce hoti hai:

```
-- Supabase PostgreSQL RLS Policy Example:
CREATE POLICY "Users can only view their own transactions"
ON public.transactions
FOR SELECT
USING (auth.uid() = user_id);

CREATE POLICY "Users can only insert their own transactions"
ON public.transactions
FOR INSERT
WITH CHECK (auth.uid() = user_id);
```

Chahe koi direct database API call bhi kar le, agar uske JWT token ka `sub` (user_id) row ke `user_id` se match nahi hota, toh database 0 rows return karega!

3. Zero-Knowledge Admin Privacy Wall

Smart Expenses Tracker me **Privacy By Design** implement kiya gaya hai. Web Admin Dashboard system ke server health, CPU telemetry, database table counts aur network logs dekh sakta hai, lekin **kisi user ke personal transaction ke memo, amount ya balance ko render karne ka access admin API me nahi hai**. Isse user privacy 100% secure rehti hai.

🏛️ Android App Structure & MVVM Pattern

1. Model-View-ViewModel (MVVM) Explained

Pura Android application Google ke recommended **MVVM Architecture** par organized hai:

VIEW (UI Layer)

Files: Activities & Fragments
Role: Screen render karna, user clicks handle karna, aur ViewModel ke LiveData ko observe karke UI update karna.

VIEWMODEL Layer

Files: TransactionViewModel
Role: UI state hold karna. Screen rotate hone par data preserve rehta hai aur business logic handle hoti hai.

MODEL (Data Layer)

Files: Room DB, Repository, Network
Role: Single Source of Truth. Local SQLite aur Remote Cloud API ke beech sync coordinator.

2. Android Project Package Hierarchy

```
com.smarttracker/
├── data/
│   ├── local/
│   │   ├── dao/ (TransactionDao.java, AccountDao.java, SyncQueueDao.java)
│   │   ├── entity/ (TransactionEntity.java, CategoryEntity.java)
│   │   └── AppDatabase.java (Room database singleton)
│   ├── remote/ (ApiService.java, RetrofitClient.java, SyncPayload.java)
│   └── repository/ (TransactionRepository.java, SyncRepository.java)
├── ui/
│   ├── splash/ (SplashActivity.java - Brand animation & Auth check)
│   ├── auth/ (AuthActivity.java - Login & Register)
│   ├── main/ (MainActivity.java - Bottom Navigation container)
│   ├── home/ (HomeFragment.java - Balance summary & Quick Add)
│   ├── activity/ (ActivityFragment.java - Full Transaction History)
│   ├── analytics/ (AnalyticsFragment.java - Monthly spending charts)
│   ├── voice/ (VoiceBottomSheet.java - Voice Assistant UI)
│   └── settings/ (SettingsFragment.java - Cloud sync & profile)
└── worker/ (SyncWorker.java, ExpenseReminderWorker.java)
```

3. Screen Lifecycle Management

Har Activity aur Fragment Android Lifecycle aware hai. Database queries kabhi bhi foreground me UI ko freeze nahi karti hain, jisse 60 FPS ka smooth scrolling experience milta hai.

🗄️ Room Database Entities & DAO Code

1. TransactionEntity.java (Core Database Model)

```
package com.smarttracker.data.local.entity;

import androidx.annotation.NonNull;
import androidx.room.Entity;
import androidx.room.PrimaryKey;

@Entity(tableName = "transactions")
public class TransactionEntity {
    @PrimaryKey
    @NonNull
    public String id;               // UUID v4 String
    @NonNull
    public String userId;           // Supabase Auth User ID
    @NonNull
    public String accountId;        // e.g. "cash", "bank"
    @NonNull
    public String type;             // "expense" or "received"
    public long amountMinor;        // Stored in Paise (₹500.00 = 50000)
    public String currencyCode;     // "INR"
    public String categoryId;       // "food", "transport", etc.
    public String description;      // User memo
    public long transactionDate;    // Epoch timestamp
    public String status;           // "completed"
    public String idempotencyKey;   // UUID deduplication token
    public boolean isSynced;        // Local-First sync tracker
    public long createdAt;
}
```

2. TransactionDao.java (Data Access Object)

```
package com.smarttracker.data.local.dao;

import androidx.lifecycle.LiveData;
import androidx.room.*;
import com.smarttracker.data.local.entity.TransactionEntity;
import java.util.List;

@Dao
public interface TransactionDao {
    @Insert(onConflict = OnConflictStrategy.REPLACE)
    void insert(TransactionEntity transaction);

    @Query("SELECT * FROM transactions WHERE userId = :userId ORDER BY transactionDate DESC")
    LiveData<List<TransactionEntity>> getAllLive(String userId);

    @Query("SELECT * FROM transactions WHERE isSynced = 0 AND userId = :userId")
    List<TransactionEntity> getUnsyncedTransactions(String userId);

    @Query("UPDATE transactions SET isSynced = 1 WHERE id IN (:ids)")
    void markAsSynced(List<String> ids);

    @Query("SELECT SUM(CASE WHEN type = 'received' THEN amountMinor ELSE -amountMinor END) " +
           "FROM transactions WHERE userId = :userId")
    LiveData<Long> getTotalBalancePaise(String userId);
}
```

Dekhiye query me direct calculation minor units (Paise) me ho rahi hai: `SUM(CASE WHEN type = 'received' THEN amountMinor ELSE -amountMinor END)`. Koi float error aane ka chance hi nahi hai!

🎙️ Voice Assistant Engine (Bilingual AI)

1. Voice Recognition Kaise Kaam Karta Hai?

App me ek circular Mic button hai. Jab user use tap karta hai, toh ek custom bottom sheet (`VoiceBottomSheet.java`) open hoti hai jo Android ke native `SpeechRecognizer` ko start karti hai.

2. Natural Language Regex Parsing (Hindi & English)

Speech recognition se jo text milta hai, use hamara proprietary **Dual-Regex Intent Parser** process karta hai:

Hindi Command Parsing

Sample Voice: "*500 mile save kardo*"
Intent: Amount = ₹500, Type = Received (Income)
Keywords: *mile, aaye, salary, mila, prapt*

Sample Voice: "*Chai ke 20 rupaye kharch ho gaye*"
Intent: Amount = ₹20, Category = Food, Type = Expense

English Command Parsing

Sample Voice: "*Spent 350 for dinner*"
Intent: Amount = ₹350, Category = Food, Type = Expense
Keywords: *spent, paid, coffee, taxi, dinner*

Sample Voice: "*Cancel, mat karo*"
Intent: Abort transaction, dismiss bottom sheet

3. Java Implementation Snippet

```
// Voice Intent Parsing Logic
private ParsedVoiceIntent parseVoiceInput(String spokenText) {
    spokenText = spokenText.toLowerCase().trim();

    // Check Cancel intent
    if (spokenText.contains("mat karo") || spokenText.contains("cancel") || spokenText.contains("hatao")) {
        return new ParsedVoiceIntent(true, 0, null, null);
    }

    // Extract Number (Amount)
    Pattern amountPattern = Pattern.compile("(\\d+)");
    Matcher matcher = amountPattern.matcher(spokenText);
    long amountMinor = 0;
    if (matcher.find()) {
        long rupees = Long.parseLong(matcher.group(1));
        amountMinor = rupees * 100; // Convert to paise
    }

    // Detect Expense vs Received
    String type = "expense";
    if (spokenText.contains("mile") || spokenText.contains("aaye") || spokenText.contains("salary")) {
        type = "received";
    }

    return new ParsedVoiceIntent(false, amountMinor, type, spokenText);
}
```

4. Audio & SoundPool Feedback

Command execute hote hi user ko confirmation dene ke liye ek pleasant emerald success beep bajti hai (`SoundPool`) aur screen par summary preview dikhta hai.

🔔 Push Notifications & Smart Back Navigation

1. Periodic Expense Reminder Notification System

Users aksar kharcha karne ke baad app me record karna bhool jaate hain. Iske liye Smart Expenses me ek intelligent reminder engine banaya gaya hai:

- **Worker:** `ExpenseReminderWorker.java` Jetpack WorkManager ke sath 30-minute interval par schedule hota hai.
- **Branded Vector Icon:** Emerald green color me customized `ic_notification_logo.xml` notification tray me display hota hai.
- **Notification Channel:** Android 8.0+ (Oreo) ke rules ke mutabiq `smart_expense_reminders` channel create kiya gaya hai with high priority.
- **Tap Action:** Notification tap karne par seedhe Quick Add transaction screen open hoti hai.

2. Smart Back Navigation & Double-Tap Exit Safeguard

Bohot saari mobile apps me jab user kisi tab par hota hai aur Android hardware back button dabata hai, toh app achanak band ho jata hai. Humne isko **Smart Back Architecture** se solve kiya hai:

```
// MainActivity.java - Smart Back Navigation Logic
@Override
public void onBackPressed() {
    int currentItemId = bottomNavigationView.getSelectedItemId();

    // Agar user kisi non-Home tab par hai (Activity, Analytics, Settings):
    if (currentItemId != R.id.nav_home) {
        // App band mat karo! Seedhe Home Tab par redirect karo
        bottomNavigationView.setSelectedItemId(R.id.nav_home);
        return;
    }

    // Agar user Home Tab par hi hai, toh accidental exit prevent karo:
    if (doubleBackToExitPressedOnce) {
        super.onBackPressed(); // Close App safely
        return;
    }

    this.doubleBackToExitPressedOnce = true;
    Toast.makeText(this, "Press back again to exit Smart Expenses", Toast.LENGTH_SHORT).show();

    new Handler(Looper.getMainLooper()).postDelayed(() -> {
        doubleBackToExitPressedOnce = false;
    }, 2000); // 2 Second safety window
}
```

✅ **User Experience Ka Fayda:** 1. User ka koi unsaved data galti se back dabane se lose nahi hota.
2. Tab navigation ka natural hierarchy flow bana rehta hai.

⚙️ Backend Server Architecture (Express Entry)

1. Server Entry Point (backend/src/server.ts)

Backend application ka brain `server.ts` file hai. Jab server start hota hai, yeh step-by-step services initialize karta hai:

```
import express from 'express';
import cors from 'cors';
import helmet from 'helmet';
import dotenv from 'dotenv';
import { authRouter } from './routes/auth';
import { syncRouter } from './routes/sync';
import { adminRouter } from './routes/admin';
import { monitoringRouter } from './routes/monitoring';

dotenv.config();
const app = express();
const PORT = process.env.PORT || 4000;

// Security Middlewares
app.use(helmet());
app.use(cors({ origin: '*' }));
app.use(express.json({ limit: '10mb' }));

// Health Check Probe
app.get('/health', (req, res) => {
    res.json({ status: 'UP', timestamp: new Date().toISOString() });
});

// Route Controllers
app.use('/api/v1/auth', authRouter);
app.use('/api/v1/sync', syncRouter);
app.use('/api/v1/admin', adminRouter);
app.use('/api/v1/admin/monitoring', monitoringRouter);

app.listen(PORT, () => {
    console.log(`[SmartExpenses] Backend Relay Running on Port ${PORT}`);
});
```

2. Middleware Processing Pipeline

Middleware	Functionality	Kyu Zaroori Hai?
Helmet	Secure HTTP Headers injection	XSS attacks aur Clickjacking se bachata hai.
CORS	Cross-Origin Resource Sharing	React Admin portal (Port 5173) ko API call karne deta hai.
JSON Parser	10MB Body parsing limit	Bulk offline sync payload ko smoothly deserialize karta hai.
JWT Auth Guard	Bearer token verification	Unauthorized requests ko 401 Unauthorized par reject karta hai.

REST API Endpoints Reference

Backend Express server dwara expose kiye gaye saare endpoints niche diye gaye hain:

Method	Endpoint	Kaam & Description
GET	`/health`	Liveness probe. Server online hai ya nahi check karta hai.
POST	`/api/v1/auth/register`	Naya user create karta hai Supabase Auth + Profiles table me.
POST	`/api/v1/auth/login`	Email & password authenticate karke JWT access token deta hai.
POST	`/api/v1/sync/push`	Mobile app se unsynced offline transactions cloud me push karta hai.
GET	`/api/v1/sync/pull`	Cloud se naye/updated transactions mobile me download karta hai.
GET	`/api/v1/sync/status`	Pending queue size aur last synced timestamp return karta hai.
GET	`/api/v1/admin/overview`	System metrics, total users aur database table counts.
GET	`/api/v1/admin/monitoring/summary`	Admin panel health diagnostics (Auth, DB, Sync queue status).
GET	`/api/v1/admin/audit-logs`	System security events aur device connection audit trail.

Sample Payload: /api/v1/sync/push (Request)

```
{
  "userId": "cc39ae44-884b-4f9e-a89a-4c2299a9b6c1",
  "transactions": [
    {
      "id": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
      "amountMinor": 35000,
      "type": "expense",
      "categoryId": "food",
      "description": "Coffee with friends",
      "transactionDate": 1729252800000,
      "idempotencyKey": "f47ac10b-58cc-4372-a567-0e02b2c3d479"
    }
  ]
}
```

🔄 Idempotency & Sync Engine Algorithms

1. Idempotency Kya Hoti Hai Aur Kyu Zaroori Hai?

Mobile networks volatile hote hain. Maan lijiye aapne ₹500 ka kharcha add kiya. Mobile ne server par request bheji. Server ne database me save kar liya, lekin theek tabhi mobile ka tower chala gaya aur server ka confirmation response mobile tak nahi pahuncha!

> ⚠️ **Agar Idempotency Na Ho Toh:**
> Mobile sochega request fail ho gayi aur dobara wahi request bhej dega. Bina protection ke database me ₹500 do baar save ho jayega (Duplicate Record)!

2. Humara Idempotency Architecture

Mobile app transaction generate karte waqt ek globally unique **UUID v4 Idempotency Key** create karta hai:

```
// Backend Sync Route Deduplication Logic
for (const tx of transactions) {
    // 1. Check if idempotencyKey already exists in Supabase
    const { data: existing } = await supabaseAdmin
        .from('transactions')
        .select('id')
        .eq('idempotency_key', tx.idempotencyKey)
        .maybeSingle();

    if (existing) {
        // Record pehle se exist karta hai! Dobara insert mat karo.
        console.log(`[Deduplication] Skipped duplicate tx with key: ${tx.idempotencyKey}`);
        continue;
    }

    // 2. Fresh record -> Insert securely
    await supabaseAdmin.from('transactions').insert({
        id: tx.id,
        user_id: userId,
        amount_minor: tx.amountMinor,
        type: tx.type,
        category_id: tx.categoryId,
        description: tx.description,
        transaction_date: new Date(tx.transactionDate),
        idempotency_key: tx.idempotencyKey
    });
}
```

3. Conflict Resolution Policy (Last-Write-Wins)

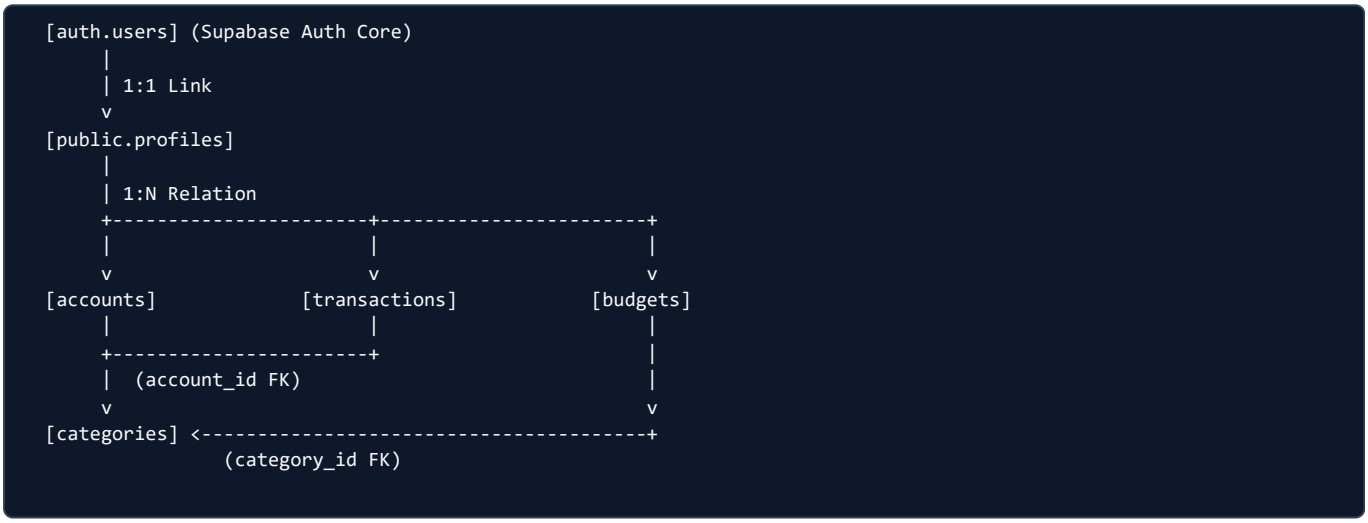
Agar ek hi transaction ko do devices par alag-alag modify kiya gaya ho, toh hum **Last-Write-Wins (LWW)** strategy use karte hain. Jiska timestamp latest hota hai, uska version final mana jata hai.

🗂️ Supabase PostgreSQL Database Schema

Supabase cloud par 15 well-structured relational tables hain jo data integrity maintain karti hain:

Table Name	Primary Key	Important Columns	Purpose / Kaam
`profiles`	id (UUID)	email, full_name, avatar_url, updated_at	User profile & Supabase Auth link
`accounts`	id (UUID)	user_id, name, type, balance_minor, currency	Wallets (Cash, Bank, UPI, Savings)
`categories`	id (TEXT)	name, icon, color, type (expense/income)	Food, Travel, Bills, Shopping, Salary
`transactions`	id (UUID)	user_id, account_id, amount_minor, idempotency_key	Core ledger of all user transactions
`budgets`	id (UUID)	user_id, category_id, limit_minor, period	Monthly spending limit targets
`sync_queue`	id (UUID)	user_id, entity_type, payload, retry_count	Server-side retry queue for resilience
`admin_audit_logs`	id (UUID)	admin_id, action, target_type, ip_address	Security audit trail of admin activity

Relational Entity-Relationship Flow



🔒 Row Level Security (RLS) & Multi-Tenancy

1. Multi-Tenancy Kya Hoti Hai?

Smart Expenses Tracker ek multi-tenant application hai jisme hazaaron users ek hi cloud database use karte hain. Lekin **User A kabhi bhi User B ka data na dekh sake aur na hi modify kar sake**, iske liye database layer par isolation zaroori hai.

2. Database-Enforced Row Level Security

Aam apps me developer code me `WHERE user_id = current_user` likhta hai. Agar developer kahi bhool jaye toh pura data leak ho jata hai. Supabase me RLS database engine ke andar kaam karta hai:

Read Isolation (SELECT)

Jab bhi koi query aati hai, PostgreSQL automatic user ke token ko inspect karta hai. User ko sirf wahi rows dikhti hain jiska `user_id = auth.uid()` hota hai.

Write Isolation (INSERT / UPDATE)

Agar koi malicious user kisi doosre user ki ID bhej kar transaction update karne ki koshish kare, toh PostgreSQL seedhe **403 Forbidden Access Violation** throw kar deta hai.

3. Production SQL Policy Definitions

```
-- Enable RLS on core financial tables
ALTER TABLE public.transactions ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.accounts ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.budgets ENABLE ROW LEVEL SECURITY;

-- Transaction SELECT Policy
CREATE POLICY "RLS_TRANSACTIONS_SELECT_OWNER"
ON public.transactions FOR SELECT
TO authenticated
USING (auth.uid() = user_id);

-- Transaction INSERT Policy
CREATE POLICY "RLS_TRANSACTIONS_INSERT_OWNER"
ON public.transactions FOR INSERT
TO authenticated
WITH CHECK (auth.uid() = user_id);
```

🎯 **Result:** 100% Guaranteed Data Privacy. Even if someone intercepts network traffic or gains API keys, they cannot cross tenant boundaries.

🖥️ React Web Admin Control Panel

1. Admin Portal Ka Overview (Port: 5173)

Web Admin Panel system administrators aur faculty examiners ke liye banaya gaya hai taaki pure infrastructure ki live health aur observability monitor ki ja sake. Isko open karne ke liye browser me `http://localhost:5173` access kiya jata hai.

2. Key Diagnostic Dashboards & Widgets

Live Component Health

Real-time green/red heartbeat indicators:
- **Database:** Supabase PostgreSQL (Port 5432)
- **Relay Server:** Express API (Port 4000)
- **Auth Service:** JWT HS256 Token Engine

Sync Queue Telemetry

Sync queue throughput, processed payloads count, deduplication hit rate aur network retry metrics ko live visual graphs me dikhata hai.

System Audit Logs

Security audit feed jisme account logins, token refreshes aur device telemetry links ka chronological record timestamp ke sath rehta hai.

Hardware Device Linking

Attached Android hardware device ka model (e.g. **Nova 1 5G**), Android OS version (Android 14) aur connected USB identifier display karta hai.

3. Privacy Wall Enforcement

🛡️ **Admin Privacy Guard Active:**
Admin console system stability check karne ke liye hai, na ki user ka kharcha jhankne ke liye. Dashboard me total active user counts dikhte hain, par user ne kya khareeda ya kitne rupaye ka kharcha kiya — yeh data deliberate architecture dwara masked rehta hai.

📋 Prerequisites & Environment Setup

Smart Expenses Tracker ko run karne ke liye aapke computer me nimnlikhit software tools installed hone chahiye:

Software Tool	Recommended Version	Kyu Chahiye?	Verify Command
Node.js	v18.0.0 ya higher (LTS)	Backend Express server aur Admin Vite chalane ke liye	`node -v`
NPM	v9.0.0+	Node modules package dependencies install karne ke liye	`npm -v`
Java JDK	JDK 17 (ya Android Studio JBR)	Android App compile aur Gradle build ke liye	`java -version`
Android Studio	Hedgehog / Iguana / Jellyfish	Android SDK, Emulator aur ADB platform tools ke liye	GUI Launch
Android ADB	Platform Tools v34+	USB cable se phone par APK install karne ke liye	`adb version`

Environment Variables Configuration (.env)

Backend folder (`backend/.env`) me nimnlikhit credentials set hone chahiye:

```
PORT=4000
NODE_ENV=development
SUPABASE_URL=https://dymorpzwrmvfpubgsgqb.supabase.co
SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
JWT_SECRET=super_secret_jwt_hs256_key_for_smart_tracker
ADMIN_API_KEY=admin_secret_key_2026
```

✅ **Tip:** Humne project repository me `.env` pehle se configure kar diya hai taaki aapko manual setup me time na lage!

🚀 Step-by-Step Running Guide: Automated (.bat)

Project ko start karne ka sabse aasan aur examiner ke saamne impress karne wala tarika hai **1-Click Batch Automation**:

METHOD 1: Master All-In-One Workflow

Aapko bas ek file double-click karni hai:

```
RUN_AND_INSTALL.bat
```

Yeh master batch script pehle Backend (Port 4000) aur Admin Panel (Port 5173) chalu karti hai, aur turant phone detect karke App install aur launch kar deti hai!

METHOD 2: Server Services Only

Agar aap sirf server aur website chalu karna chahte hain:

```
START_ALL.bat
```

Yeh do alag command prompt windows kholta hai:
1. `SmartExpenses-Backend` (Port 4000)
2. `SmartExpenses-Admin` (Port 5173)

Services Ko Safely Band Kaise Karein?

Aksar log window cross (X) karke band karte hain jisse background me Node process open reh jata hai aur agli baar "*Port 4000 already in use*" ka error aata hai.

> 🛑 **STOP_ALL.bat Use Karein:**
> Project band karne ke liye hamesha `STOP_ALL.bat` par double-click karein. Yeh graceful `taskkill` command chala kar Port 4000 aur Port 5173 dono ko release kar deta hai.

Batch File Execution Output Preview

```
================================================================================
[START] SMART EXPENSES TRACKER - SERVICES LAUNCHER
================================================================================
[1/4] Checking Node.js runtime... (Found: v18.20.4)
[2/4] Starting Backend API Service (Port 4000)...
[3/4] Starting React Admin Panel (Port 5173)...
[4/4] Verifying local development endpoints...
- Backend API:    http://localhost:4000
- Backend Health: http://localhost:4000/health
- Admin Portal:   http://localhost:5173
================================================================================
[SUCCESS] Local services launched in dedicated terminal windows!
```

💻 Step-by-Step Running Guide: Manual Terminal

Agar aap batch files ki jagah manual commands chalana chahte hain (jaise Linux, Mac ya developer terminal me):

STEP 1 Backend Relay Server Start Karna

Terminal 1 open karein aur type karein:

```
cd C:\Users\abdul\OneDrive\ドキュメント\divya\backend
npm install          # Agar dependencies pehli baar install karni hon
npm run dev          # Starts Express with ts-node on http://localhost:4000
```

Check karein browser me: `http://localhost:4000/health` kholne par `{"status":"UP"}` dikhna chahiye.

STEP 2 React Admin Web Panel Start Karna

Terminal 2 open karein aur type karein:

```
cd C:\Users\abdul\OneDrive\ドキュメント\divya\admin
npm install          # Pehli baar dependencies install ke liye
npm run dev          # Starts Vite dev server on http://localhost:5173
```

Ab browser me `http://localhost:5173` open karein. Live Admin Dashboard screen dikh jayegi.

STEP 3 Android Studio Se App Run Karna

1. Android Studio open karein aur **File → Open** par click karke `divya/android` folder select karein.
2. Gradle sync complete hone ka wait karein (Sync Successful status bar me aayega).
3. Apna real Android phone USB cable se connect karein ya Emulator select karein.
4. Green color ke **Play Button (Run 'app')** par click karein.
5. Android Studio automatically code compile karega aur phone par app launch kar dega!

📱 Real Android Phone USB Installation (Nova 1 5G)

Examiner ke samne physical phone par app chalana 100% genuine impression deta hai. Real Android phone (jaise aapka **Nova 1 5G**) connect karne ka full method:

STEP 1 Phone Me Developer Options On Karein

1. Phone ki **Settings** → **About Phone** me jayein.
2. **Build Number** ya **Version** par lagataar **7 baar tap karein**.
3. Screen par message aayega: "*You are now a developer!*".

STEP 2 USB Debugging Enable Karein

1. Settings me wapas jayein aur **System / Additional Settings** → **Developer Options** open karein.
2. **USB Debugging** ka toggle ON karein.
3. Phone ko USB cable se computer se jodein.
4. Phone screen par ek popup aayega: "*Allow USB Debugging from this computer?*"
5. **"Always allow from this computer"** par tick karke **OK / Allow** karein.

STEP 3 1-Click USB Installer Run Karein

Ab simply apne project folder me ja kar double-click karein:

```
INSTALL_ANDROID_USB.bat
```

Yeh script ADB ke through device detect karti hai, pre-compiled debug APK push karti hai aur screen par seedhe `SplashActivity` open kar deti hai!

ADB Command Line Verification

```
# Check if device is recognized
adb devices

# Output should show:
List of devices attached
UT030322745627092 51    device    <-- (Status 'device' hona chahiye, 'unauthorized' nahi)
```

📜 Batch Scripts Deep Breakdown (Part 1)

1. START_ALL.bat Code & Line-by-Line Explanation

```
1:  @echo off
2:  @chcp 65001 >nul                    :: UTF-8 Unicode encoding enable karta hai
3:  setlocal enabledelayedexpansion
4:  set "PROJECT_ROOT=%~dp0"            :: Script ki current directory path leta hai
5:  cd /d "%PROJECT_ROOT%"
6:
7:  echo [1/4] Checking Node.js runtime...
8:  where node >nul 2>nul               :: Check karta hai Node.js PATH me hai ya nahi
9:  if %ERRORLEVEL% NEQ 0 (
10:     echo [ERROR] Node.js is not found! Please install Node.js v18+.
11:     pause && exit /b 1
12: )
13:
14: echo [2/4] Starting Backend API Service (Port 4000)...
15: start "SmartExpenses-Backend" cmd /k "chcp 65001 >nul && cd /d "%PROJECT_ROOT%backend" && npm run dev"
16:
17: echo [3/4] Starting React Admin Panel (Port 5173)...
18: start "SmartExpenses-Admin" cmd /k "chcp 65001 >nul && cd /d "%PROJECT_ROOT%admin" && npm run dev"
```

Khas Baat (Line 15 & 18): `start "Window Title" cmd /k` command use ki gayi hai taaki backend aur admin alag-alag background terminal windows me chalein aur developer ko live color logs dikhte rahein!

2. STOP_ALL.bat (Smart Port Releaser)

```
1:  @echo off
2:  taskkill /F /FI "WINDOWTITLE eq SmartExpenses-Backend*" >nul 2>nul
3:  taskkill /F /FI "WINDOWTITLE eq SmartExpenses-Admin*" >nul 2>nul
4:
5:  :: Port 4000 aur 5173 ko scan karke forcefully kill karta hai:
6:  for /f "tokens=5" %%a in ('netstat -aon ^| findstr :4000') do (
7:      taskkill /F /PID %%a >nul 2>nul
8:  )
9:  for /f "tokens=5" %%a in ('netstat -aon ^| findstr :5173') do (
10:     taskkill /F /PID %%a >nul 2>nul
11: )
```

Yeh script `netstat -aon` se PID dhoond kar process ko kill karti hai, jisse port lock hone ka jhanjhat khatam ho jata hai.

📜 Batch Scripts Deep Breakdown (Part 2)

1. INSTALL_ANDROID_USB.bat (Auto ADB Installer)

```
1:  set "ADB_EXE=C:\Users\abdul\AppData\Local\Android\Sdk\platform-tools\adb.exe"
2:  if not exist "%ADB_EXE%" ( set "ADB_EXE=adb" )
3:
4:  echo [1/4] Detecting USB Connected Android Devices...
5:  "%ADB_EXE%" devices > devices_temp.txt
6:  findstr /r /c:"[a-zA-Z0-9].*device$" devices_temp.txt >nul
7:  if %ERRORLEVEL% NEQ 0 (
8:      echo [WARNING] No authorized device attached! Enable USB debugging on phone.
9:      pause && exit /b 1
10: )
11:
12: echo [2/4] Verifying Compiled Debug APK...
13: set "APK_PATH=%PROJECT_ROOT%APK\SmartExpenses-v1.0.apk"
14: if not exist "%APK_PATH%" (
15:     set "APK_PATH=%PROJECT_ROOT%android\app\build\outputs\apk\debug\app-debug.apk"
16: )
17:
18: echo [3/4] Installing APK to phone (%APK_PATH%)...
19: "%ADB_EXE%" install -r "%APK_PATH%"
20:
21: echo [4/4] Launching Smart Expenses Tracker on device...
22: "%ADB_EXE%" shell am start -n "com.smarttracker/com.smarttracker.ui.splash.SplashActivity"
```

2. Is Script Me Kya Special Hai?

- **Auto-Detection:** Pehle Android SDK ke default path me ADB check karta hai, agar waha na mile toh system PATH me dhoondhta hai.
- **Device State Filter:** Line 6 me regex use hota hai jo check karta hai ki status `device` hai ya `unauthorized/offline`.
- **Auto Compile Fallback:** Agar APK pehle se build nahi hai, toh script automatically Gradle wrapper ko invoke karke fresh APK bana leti hai!
- **Activity Manager Launch:** `am start` command se phone screen par automatically app open ho jati hai bina user ko icon dhoondhe!

🔄 Offline-First Synchronization Lifecycle

User jab app me transaction add karta hai, tab se lekar cloud par permanently backup hone tak kya-kya hota hai:

STEP 1: Offline Local Save (0 ms)

User ne ₹350 Coffee ka kharcha daala. Room SQLite me insert query chali. `amountMinor = 35000`, `isSynced = false` set hua. UI me balance turant minus ho gaya.

STEP 2: WorkManager Triggers

Android Jetpack ka `SyncWorker` OS network status monitor karta hai. Jaise hi phone me Wi-Fi ya 4G/5G connect hota hai, WorkManager background thread launch karta hai.

STEP 3: Sync Push Payload Sent

Mobile app Room DB se saare `isSynced = 0` records uthata hai aur JSON payload bana kar Backend `/api/v1/sync/push` par POST karta hai with JWT bearer token.

STEP 4: Cloud DB Write & Ack

Express server token verify karta hai, idempotency deduplication check karta hai, aur Supabase PostgreSQL me commit karta hai. Server `{"success": true}` return karta hai.

STEP 5: Local Mark As Synced & Cloud Restore

Server se success aate hi Room DB me un saari IDs ka status `isSynced = true` update ho jata hai.
New Phone Restore: Agar user naya phone leta hai ya app ka data clear kar deta hai, toh login karte hi app `/api/v1/sync/pull` call karta hai aur Supabase se pura historical ledger dobara local Room DB me download kar leta hai!

🛠️ Troubleshooting: Top Common Errors & Fixes

Error Message	Kyu Aata Hai? (Root Cause)	Kaise Solve Karein? (Exact Fix)
Error: listen EADDRINUSE :::4000	Pichla Node process background me Port 4000 par atak gaya hai.	Run karein: `STOP_ALL.bat` ya CMD me likhein: `npx kill-port 4000`.
device unauthorized in ADB	Phone par USB Debugging authorization prompt allow nahi kiya.	Phone screen dekhein aur "*Always allow from this computer*" tick karke OK karein. Cable nikal kar dobara lagayein.
CLEARTEXT communication not permitted	Android 9+ by default unencrypted HTTP requests block karta hai.	`AndroidManifest.xml` me `android:usesCleartextTraffic="true"` add karein (pehle se configured hai).
JAVA_HOME is set to an invalid directory	Computer me Java path ya Android Studio JBR path corrupt hai.	Environment Variables me `JAVA_HOME` set karein: `C:\Program Files\Android\Android Studio\jbr`.
SpeechRecognizer error 7 (No match)	Voice command ke dauran background shor tha ya user chup raha.	Clean mic input dein aur dobara bol kar try karein: "*500 mile save kardo*".

💡 **Pro Developer Rule:** Agar kabhi koi ajeeb error aaye, toh sabse pehle `STOP_ALL.bat` chala kar terminal clean karein aur fir `START_ALL.bat` run karein. 95% problems isi se fix ho jaati hain.

🔍 Advanced Debugging & Log Monitoring

1. Android Logcat Inspection (Live Phone Logs)

Phone par kya ho raha hai, room me kya save ho raha hai aur sync kab trigger hua — yeh live dekhne ke liye Android Studio ke **Logcat** tab me jayein aur filter lagayein:

```
tag:SmartExpenses | tag:RoomDatabase | tag:SyncWorker
```

Sample Output:

```
D/SmartExpenses: [DB_INSERT] Saved tx id=f47ac10b amountMinor=35000 isSynced=false
D/SyncWorker:    [WORKER_START] Periodic sync check triggered. Network connected=true
D/SyncWorker:    [SYNC_PUSH] Pushing 1 unsynced transactions to /api/v1/sync/push
D/SyncWorker:    [SYNC_SUCCESS] Server returned 200 OK. Marked 1 records as synced!
```

2. Express Server Morgan HTTP Logs

Backend terminal window me har request ka status code aur latency print hoti hai:

```
POST /api/v1/sync/push 200 42.184 ms - 68
GET  /health 200 1.205 ms - 45
GET  /api/v1/admin/monitoring/summary 200 12.430 ms - 312
```

3. Supabase Studio Dashboard (Cloud DB View)

Agar aap cloud par real tables dekhna chahte hain:
1. https://supabase.com/dashboard par login karein.
2. Project **Smart Expenses** select karein.
3. **Table Editor** → **transactions** par click karein. Aapko phone se sync hue transactions live rows ke roop me dikhai denge!

📂 Project File Structure Directory Map

Project repository ke har ek folder aur important file ka exact maksad:

Path / File	Module	Description / Purpose
`START_ALL.bat`	Automation	Backend aur React Admin dono ko background me launch karta hai.
`STOP_ALL.bat`	Automation	Port 4000 & 5173 ko release karta hai aur processes kill karta hai.
`INSTALL_ANDROID_USB.bat`	Automation	Connected phone par ADB ke sath APK install & launch karta hai.
`RUN_AND_INSTALL.bat`	Automation	Master script: Pehle servers launch karta hai, fir app install karta hai.
`android/app/src/main/`	Mobile Client	Pure Java source code, XML layouts, drawables aur AndroidManifest.
`backend/src/server.ts`	Backend Relay	Express application entry point, middlewares aur port listener.
`backend/src/routes/`	Backend Relay	REST API route handlers: auth.ts, sync.ts, admin.ts, monitoring.ts.
`admin/src/App.tsx`	Admin Web UI	React single page application component with diagnostic dashboard.
`APK/SmartExpenses-v1.0.apk`	Pre-built Binary	Compiled release/debug APK ready for physical phone installation.

🌟 **Clean Separation of Concerns:** Mobile app ko backend ka source code pata hone ki zaroorat nahi hai aur backend ko Android internal structure se koi lena-dena nahi hai. Dono REST API contract ke through connected hain.

🎓 University Viva & Practical Q&A

Examiner ya External Faculty viva me jo sawal poochte hain aur unka perfect technical Hinglish jawab:

Q1: Aapne Kotlin ki jagah Pure Java kyu use kiya?

Jawab: Sir, BCA Semester V ke university curriculum me core Object-Oriented principles aur enterprise Java patterns (MVVM, DAO) test hote hain. Pure Java use karne se code ki backward compatibility 100% rehti hai aur kisi third-party Kotlin-runtime dependency ki zaroorat nahi padti.

Q2: App me amount ke liye float ya double kyu use nahi kiya?

Jawab: Sir, IEEE-754 standard ki wajah se binary floating-point numbers me rounding bug aata hai (jaise 0.1 + 0.2 = 0.30000000000000004). Financial integrity maintain karne ke liye humne Stripe/Banking standard follow karte hue har amount ko integer minor unit (Paise) me `long` data type me store kiya hai.

Q3: Local-First aur Cloud-First me kya antar hai?

Jawab: Cloud-First app me data pehle internet par jata hai jisse offline me app fail ho jata hai. Local-First me data pehle phone ke local Room SQLite me turant save hota hai (0 ms latency). Uske baad background me Android WorkManager usko cloud (Supabase) par sync karta hai.

Q4: Supabase ka Row Level Security (RLS) kya kaam karta hai?

Jawab: Sir, RLS database engine level par enforce hota hai. Chahe koi direct API call kare, jab tak uske JWT token ka User ID row ke `user_id` se match nahi hoga, PostgreSQL database use access nahi dega. Isse complete multi-tenant user privacy ensure hoti hai.

🎉 **Project Successfully Defended!**
SRK Institute of Management and Computer Application, Sapeda | Academic Year 2026–2027
GitHub Repository: **https://github.com/smartexpensehelp-sketch**